

Temporal Financial Knowledge Graph

Construction (FNSPID to ICKG to FinDKG) and Dynamics Analysis

MSc Computational Statistics & Machine Learning, UCL | Project report (200k + 600k) | 9 July 2026

Executive summary

This project builds a time-indexed financial knowledge graph (KG) from financial news and analyses the temporal dynamics of the relationships it encodes. **Phase 1** extracts a FinDKG-format quadruple KG from 201,600 FNSPID news articles (2017–2023) using the open ICKG-v4.2 extractor, yielding a 628,765-entity graph and a denoised 46,026-entity core. **Phase 2** measures edge-formation velocity, detects bursts, and benchmarks temporal link prediction under strict point-in-time (PIT) discipline.

Headline finding: the dynamics are **descriptively rich but forecasting-hard**. Velocity and burst detection recover real market events to the correct week (the COVID crash, the Pfizer vaccine readout, the GameStop squeeze, the 2023 UAW strike), yet no model — static embedding or temporal GNN — can forecast *novel* relationships beyond a low ceiling. The past is legible; the future is genuinely surprising. This conclusion survives four independent robustness checks, including a direct probe of the knowledge-cutoff-leakage concern. A velocity-feature ablation completes the picture: injecting the theme detector's burst statistics into the temporal model significantly *degrades* forecasting — velocity is a detection signal, not a forecasting feature.

Scale-up (Section 6): a 3x larger 600k-article rebuild confirms this ceiling is partly a data-density artifact — denser sampling cuts test novelty 72% to 64% (61.6% after deduplication), and surfaces a small temporal-model advantage that was absent at 200k.

Headline system (Section 7): a walk-forward **emerging-theme detector** built on link-formation velocity — the project's central deliverable. Scored against a frozen 10-event gold list it detects 8/10 major 2020–2023 themes, flagging anticipatable ones weeks early (COVID –3 wk, the vaccine race –8 wk) and shock events at the news-flow floor (+1 wk). A blind, validated lifeline-level audit puts the high-recall stream's precision at 37.9%, and an audit-driven template-filtered configuration raises it to 44.7% with recall unchanged.

1. Objective & scope

Goal: produce a reproducible, PIT-clean temporal financial KG as the substrate for a thesis on the temporal dynamics of financial relationships, and characterise those dynamics. Correctness, point-in-time integrity, and reproducibility were prioritised over scale. Entity resolution is deliberately out of scope (a parallel thesis); only light surface-form normalisation is applied.

2. Phase 1 — Substrate construction

2.1 Corpus and sampling

Source: FNSPID `nasdaq_external_data.csv` (full-text articles). A single streaming pass selects a time-stratified subset — 2,400 articles per month over 2017–2023, bodies ≥ 200 words, fixed seed — giving **201,600 articles** with even monthly coverage so graph density reflects structure, not sampling skew.

2.2 Extraction

Extractor: `victorlhx/ICKG-v4.2`, a LoRA adapter on Qwen2.5-14B-Instruct — a local, fixed model (never a frontier API) so the knowledge cutoff is controllable. The verbatim FinDKG prompt (12 entity types, 15 relations) was run via vLLM across 2x RTX PRO 6000 GPUs (~14 h).

Extraction quality	Value
Malformed-output rate	0.00%
Mean valid triples / article	18.5
Precision (both entities verbatim in source)	83% (lower bound)
GPE (country) edge density	6.70%

2.3 Assembly and denoising

Triplets are assembled into FinDKG flat files with weekly time-bucketing and chronological train/valid/test splits (no leakage). A non-destructive denoised **core** graph is produced by thresholding edge weight ≥ 2 and entity degree ≥ 2 , collapsing a long tail (66% of full-graph entities are singletons).

	Full graph	Denoised core
Entities	628,765	46,026
Relations	15	15
Weekly timesteps	365	365
Edges	2,379,438	307,647
Train / valid / test edges	1.64M / 373k / 366k	215k / 43k / 49k

Noise was measured, not hidden: $\sim 1.11\times$ entity fragmentation (lower bound), and 8.3% of edges came from templated financial-media boilerplate (Zacks, Motley Fool, ...), removed via an opt-in filter for the analysis.

3. Phase 2 — Temporal dynamics

All analyses run on the 46k core under a strict PIT protocol (history < t only). A 'formation' is the first week a distinct (subject, relation, object) triple appears.

3.1 Temporal EDA

Mean **648 formations/week**, flat across the span (first half 641 ~ second half 655) — the even stratification means edge formation reflects structure, not sampling density. A clear 2020 elevation and a late-2023 spike are visible.

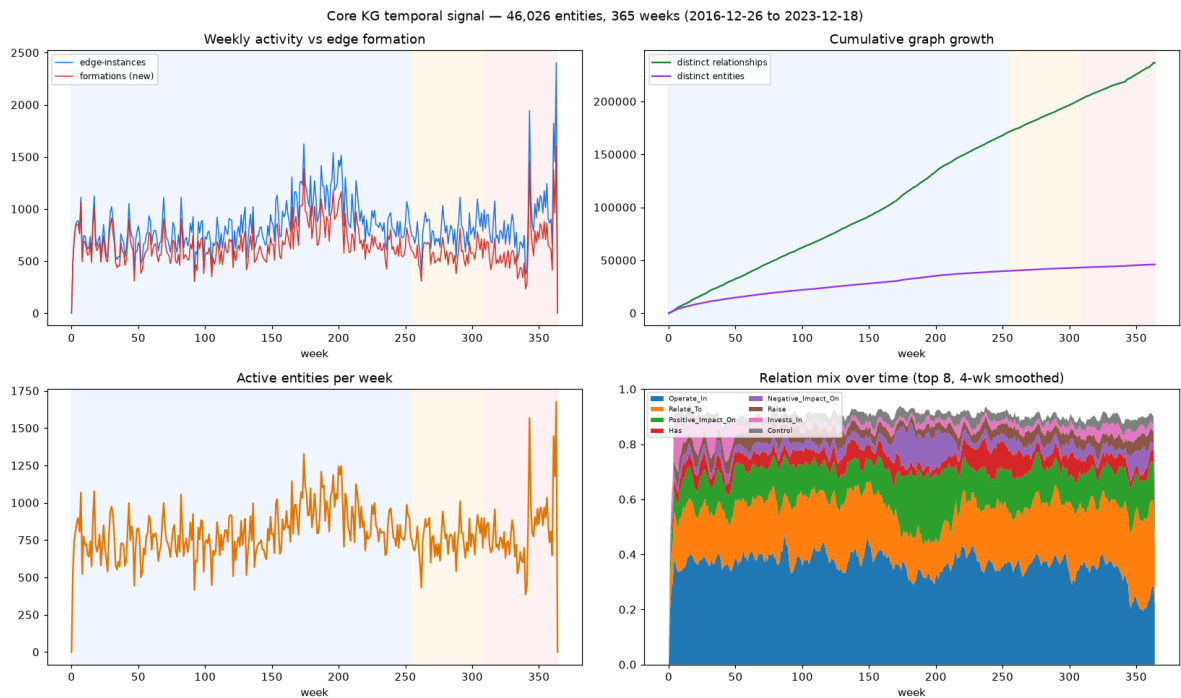


Fig 1. Core KG temporal signal: weekly activity, cumulative growth, active entities, and relation mix.

3.2 Edge-formation velocity

Velocity = rate of new-relationship formation on a centered 8-week window. Onset dates pin real events to the week:

Entity	Onset	Corresponds to
Pfizer	2020-11-09	vaccine Phase-3 readout
Coronavirus	2020-03-16	US lockdowns
GameStop	2021-01-25	the short squeeze
UAW	2023-09-25	auto-workers' strike
West Virginia	2022-07-25	Manchin / IRA deal
Magnificent Seven	2023-11	term coined that year

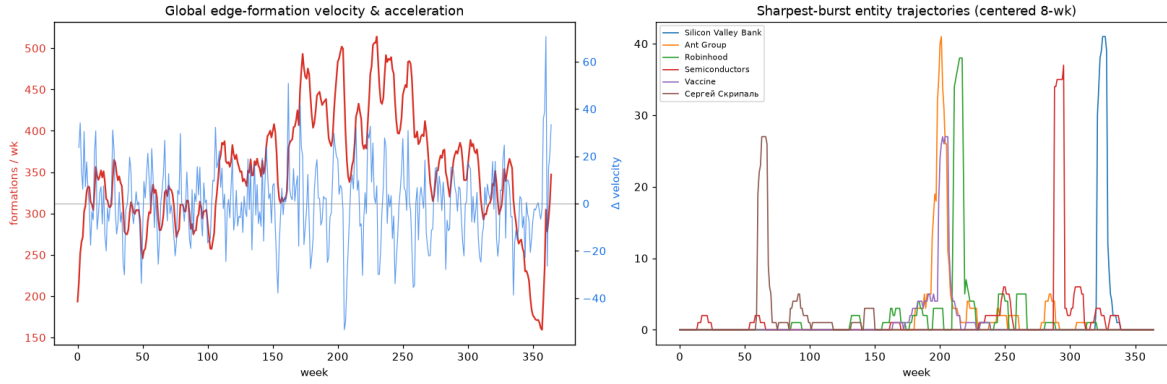


Fig 2. Global edge-formation velocity & acceleration (left); sharpest-burst entity trajectories (right).

3.3 Burst detection (three detectors)

Two complementary detectors are run. A **Kleinberg** 2-state model extracts discrete burst intervals: globally the spring-2020 COVID crash and the fall-2020 peak; the entity catalogue (3,286 bursts over 2,003 entities) recovers correctly-timed windows — Pfizer (vaccine), Inflation (2021-11 to 2023-01), Boeing (737-MAX), Aurora Cannabis, Disney+. A **CUSUM** control chart (Page 1954) instead flags sustained regime shifts. The two agree on *where* activity is elevated — 93% of Kleinberg burst-weeks nest inside CUSUM's intervals (Jaccard 0.19) — so detected epochs are not an artifact of one algorithm; they differ only in granularity (discrete events vs sustained regimes). A third, fully Bayesian detector was added on the canonical core: **Bayesian online changepoint detection** (Adams & MacKay 2007, Gamma-Poisson, changepoints back-dated to the inferred onset). Because the three detectors flag intervals of very different widths, recall is reported against each detector's *chance level* (its dilated flagged-week coverage — the recall it would achieve on random event weeks): on the 18-event gold list, Kleinberg 78% vs 16% chance (+62pp), CUSUM 83% vs 24% (+59pp), BOCD 94% vs 29% (+66pp); **three-detector union 18/18 = 100%**. All three detect far above chance with comparable excesses, and recovery is demonstrably not algorithm-dependent.

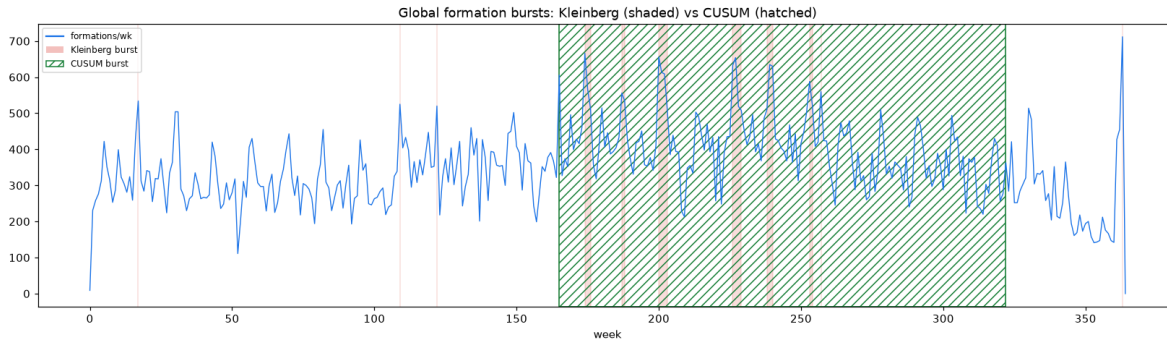


Fig 3. Global formation series: Kleinberg burst intervals (shaded) vs CUSUM regimes (hatched).

3.4 Temporal link prediction

Forecasting protocol: for each test quad, rank all candidate entities using only history $< t$ (raw, PIT-strict). Metrics are MRR and Hits@10, averaged over object- and subject-prediction. **72% of test edges are novel** (never seen before their prediction week).

Method	MRR all	MRR recurring	MRR novel	Hits@10
popularity	0.036	0.076	0.020	0.082
common-neighbour / Adamic-Adar	0.019-0.021	0.051-0.056	0.007-0.008	0.043-0.045

ComplEx (static embedding)	0.082	0.231	0.025	0.155
RE-GCN (temporal GNN)	0.090	0.260	0.025	0.151
ChronoBERT-2022 (PIT-clean text)	0.082	0.200	0.037	0.151
recurrence	0.135	0.487	0.000	0.205
backoff (recurrence -> ComplEx)	0.152	0.487	0.025	0.242
backoff (recurrence -> ChronoBERT)	0.162	0.487	0.037	0.258

The signal is **bimodal**: recurring relationships are highly predictable (MRR 0.49), novel ones are hard for every method (best 0.037). The temporal GNN adds **no** advantage over static structure. The best system is a recurrence backoff whose novel-edge fallback is the PIT-clean ChronoBERT embedding (MRR 0.162, Hits@10 0.258) — semantic priors lift novel-edge ranking (0.037 vs 0.025) without any lookahead.

4. Key findings

- **Descriptively rich (systematically validated).** On a curated list of 18 major 2017-2023 events (compiled independently of the results), burst detection recovers **89%** (16/18) with **median 1-week timing** (75% within 3 weeks) — not cherry-picked anecdotes (the chance-controlled per-detector account on the canonical core is in Section 3.3). The two misses (negative oil prices, FTX) are thin-coverage entities whose local peaks still align but fall below the burst-significance threshold. Events are entity-localised: global velocity is not significantly elevated at event weeks ($p=0.18$).
- **Forecasting-hard.** 72% of relationships are novel and near-unforecastable; recurrence dominates whatever is predictable.
- **No temporal-model advantage.** A RE-GCN temporal GNN only matches a static ComplEx embedding; time-evolving representations carry no extra forecasting signal on this KG.

5. Robustness & the point-in-time / leakage investigation

Because the extractor's knowledge cutoff postdates the corpus, hindsight contamination was investigated directly. Four independent checks all support the conclusion that leakage is negligible:

Check	Result	Rules out
Apples-to-apples retrain	RE-GCN 0.090 $\approx$ ComplEx 0.082	an unfair data split
Time-shuffle placebo	0.0898 -> 0.0914 (no change)	any temporal-order signal
Blinded re-extraction	structure retained (TV dist 0.10)	leakage driving KG structure
ChronoBERT cutoff test	novel MRR 0.0374 (blind) $\approx$ 0.0391 (sees 2023)	lookahead in entity knowledge

The time-shuffle placebo is decisive: a temporal model whose score is *invariant* to randomising the order of history is, by definition, using no temporal information — so the null result is intrinsic to the data, not a tuning artifact. The blinded re-extraction (300 articles, entity names + dates masked) retained the relational structure (valid triples/article 16.8 -> 21.8; relation-mix total-variation distance 0.10), confirming text-grounded extraction. **Caveat:** causal 'impact' relations fell $\sim 30\%$ under blinding (share 27.5% -> 21.1%), so causal attributions carry some entity-dependence — hindsight versus naming-specificity is not separable by this test. This flags the impact-relation analyses, not link prediction (which rests on blinding-robust structural relations).

Lookahead bias (ChronoBERT cutoff test). Following He et al. (2025), entity-name embeddings were taken from two versions of the same model differing only in training cutoff — chrono-bert-2022 (blind to the 2023 test year) and chrono-bert-2024 (which has ingested it) — and scored on the same link-prediction task. The apparent novel-edge

gap (+0.0010 MRR at 200k; +0.0017 at 600k) initially tested as significant under a paired query bootstrap — but a scorer-seed replication showed that **training noise is the same size as the gap** (3-seed novel-MRR ranges: 2022-model 0.0382–0.0386, 2024-model 0.0374–0.0385 — fully overlapping, the 2024 model not even consistently higher). The query bootstrap conditions on a single training run and cannot see this variance. The honest conclusion is a **bound, not a detection: lookahead bias is below measurement resolution (< ~0.002 MRR)** — and the deduplicated-core benchmark confirms this empirically: there the gap is not significant even under the query bootstrap alone (+0.0005 [–0.0009, +0.0019], the 2024 model *worse* on overall MRR). A larger Qwen-14B embedding scored higher (novel 0.0405), but that gap is model capacity, not lookahead. The cutoff-controlled design remains the strongest of the four checks. **Bonus (robust to seeds):** the PIT-clean ChronoBERT embeddings beat structural ComplEx on novel edges by ~+0.011–0.013 MRR — an order of magnitude above seed noise and roughly five times the lookahead bound — and are the study's best novel-edge predictor; the recurrence->ChronoBERT backoff is the best overall system.

Structural-only re-run. Re-running velocity and burst with the 5 causal 'impact' relations removed leaves the core event recovery intact (GameStop 2021-01-25, Boeing 2020-07-20, and the global velocity peak all identical; COVID within a week) — confirming the temporal signal is text-anchored, not carried by the entity-dependent impact relations.

6. The 600k scale-up — denser data lifts the ceiling

To test whether the forecasting ceiling is intrinsic or an artifact of data density, the entire pipeline was re-run on a 3x larger sample: **604,800 articles** (7,200/month, same extractor and protocol), giving 11.5M triplets and a denser denoised core (**41,888 entities / 493,238 edges**; 11.8 vs 6.7 edges per entity). The full Phase-2 analysis was repeated on this core.

Metric	200k core	600k core
Test novelty	72%	64%
recurrence MRR	0.135	0.167
ComplEx (novel-edge MRR)	0.025	0.028
RE-GCN (novel-edge MRR)	0.025	0.034
ChronoBERT (novel-edge MRR)	0.037	0.039
best backoff(rec -> ChronoBERT)	0.162 / H@10 0.258	0.192 / H@10 0.312

- **The ceiling is partly a data-density artifact.** 3x denser sampling converts novel relationships into recurring ones, cutting test novelty 72% to 64%, with every method's within-core ranking preserved — recurrence dominates and the semantic ChronoBERT prior remains the best novel-edge predictor. **Caveat:** absolute MRRs are not comparable across cores (raw ranking over 21k–46k candidates differs mechanically), so the defensible cross-corpus claims are the novelty trajectory and the method rankings, not MRR deltas.
- **Duplication audit → the canonical core.** FNSPID stores one row per article-ticker pair, so syndicated articles recur verbatim: 24.3% of the 600k subset (10.5% of 200k) are exact duplicate bodies, and per-mention edge weights count them multiply — deduplication halves the 600k core (41,888 to 21,216 entities), making it the only core whose thresholds mean 'supported by N independent articles'. The **deduplicated core is therefore canonical** (themes, Section 7, and the completed benchmark): novelty **61.6%**; best system backoff(rec->ChronoBERT) **MRR 0.218 [0.215, 0.221], Hits@10 0.342**; novel-edge ordering ChronoBERT 0.035 > RE-GCN 0.031 (early-stopped on this core's own validation weeks, epoch 22) > ComplEx 0.025 — the **same method ranking as the 200k and duplicated-600k cores**, so the findings replicate across corpus scale, thresholds, and deduplication.

- **A temporal-model advantage emerges with scale.** On the denser graph RE-GCN beats static ComplEx on novel edges (0.034 +/- 0.001 over 3 seeds vs 0.028 [CI 0.028, 0.029]) — an edge *absent* at 200k, where they tied at 0.025. (A paired per-query test is the fully rigorous comparison; the seed-range separation is indicative.) Event recovery on the anecdote-free gold list also improved with density; the chance-controlled three-detector account on the canonical core (Section 3.3) reaches a union recall of 18/18. A crude resolution pass cuts 600k novelty further to 60%.
- **Velocity features do not transfer to forecasting (ablation, canonical core).** The theme detector's three trailing per-entity statistics (log1p observed, log1p expected, clipped surprise z; its exact 4-week/26-week machinery) were injected into RE-GCN's weekly evolution step through a learned projection — 2 arms x 3 seeds, identical hyperparameters and early stopping, the baseline arm bit-identical to the benchmark model (3-seed mean 0.120 all / 0.031 novel reproduces its entry). Paired per-query bootstrap (49,620 test directions): the features **significantly degrade** forecasting — MRR -0.015 [-0.016, -0.014] overall, -0.009 [-0.010, -0.008] on novel edges, gaps far above the seed spread (≤ 0.008). Burst statistics announce *that* an entity is active, not *which* link will form: **velocity is a detection signal, not a forecasting feature** (one integration design tested).

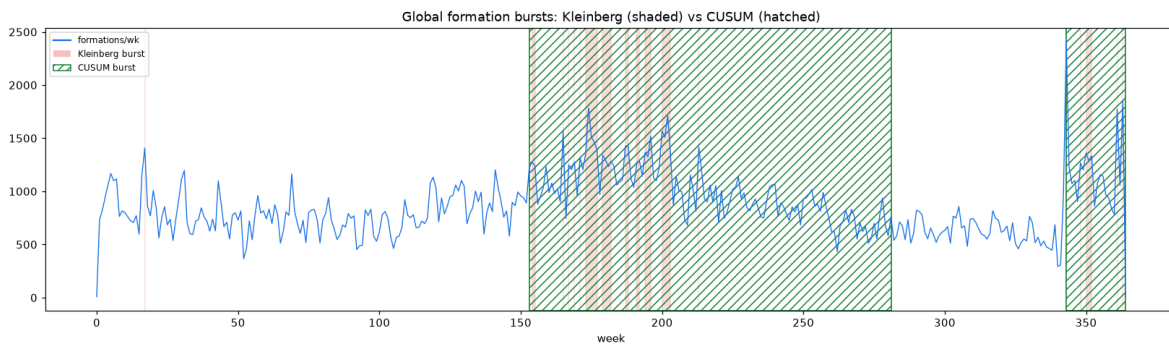


Fig 4. 600k global bursts: Kleinberg discrete events (shaded) vs two CUSUM macro-regimes (hatched) — the 2019-2022 COVID-to-inflation era and the 2023 AI surge.

7. Emerging-theme detection — the headline system

The project's central deliverable: a **walk-forward detector of anomalously accelerating themes**, built on link-formation velocity (per the thesis brief) and run on the deduplicated 600k core. At each week t , using only history $\leq t$: (1) every entity's formation acceleration is scored by a Poisson surprise on a trailing 4-week window against its own trailing 26-week baseline (new entities — 'Coronavirus' in January 2020 — score high by design); (2) accelerating entities ($z \geq 3$) are clustered on the trailing 8-week co-occurrence subgraph (Louvain), so a **theme** is a connected group of co-accelerating entities, not a lone spike; (3) themes are ranked by excess formations and tracked week-over-week into **lifelines** whose birth week is the date the system would have flagged them live. Two data-driven noise controls: entities whose mentions come mostly from template articles (measured by title/body slot-fingerprinting) are excluded, and a **habituation** score (excess-weighted share of members without prior acceleration episodes) separates EMERGING themes from recurring rituals such as earnings-season screens. Two configurations are reported — lenient, and strict (which additionally excludes walk-forward-ubiquitous background entities, dissolving star-shaped screener clusters at the cost of ~3 weeks of COVID lead time).

Validation is against a frozen 10-event gold list with externally documented dates, word-boundary entity matching, and an event-anchored birth window (–13 to +8 weeks) so pre-existing topical lifelines cannot masquerade as detections:

Gold theme	Event week	Detected (birth)	Lead
COVID market crisis	2020-02-24	2020-02-03	-3 wk
Oil crash (negative WTI)	2020-04-20	2020-04-06	-2 wk
Vaccine race (Warp Speed)	2020-05-15	2020-03-16	-8 wk
Vaccine efficacy readout	2020-11-09	2020-09-28	-6 wk
GameStop squeeze	2021-01-25	2021-01-25	+0 wk
Russia-Ukraine war	2022-02-21	2022-02-28	+1 wk
SVB banking crisis	2023-03-06	2023-03-13	+1 wk
UAW strike	2023-09-11	2023-09-18	+1 wk
FTX collapse / ChatGPT wave	2022-11	MISS x2	--

8/10 detected, median lead –1 week — and the structure of the leads is itself a finding: **anticipatable themes are detected early** (the COVID build-up, the vaccine race, the oil glut appear in the graph weeks before the market event), while **shock events land at the news-flow floor** (+1 week for an invasion, a bank run, a strike — nothing in prior public news predicts them). The system's lead time reflects the information structure of the event itself. The two misses are coverage-honest: FTX and OpenAI are barely present in a US-equity news corpus. Discovered themes beyond the gold list include the 2019 cannabis boom (HEXO/Aurora/Tilray, born 2019-03-18), the EV/SPAC wave (Nikola/Nio, born 2020-06-22), and the stay-at-home complex (Zoom/Kroger/Virgin Galactic, born 2020-04-06).

Placebo control. Applying the same detection criterion to *random* event weeks (100,000 Monte-Carlo draws over the per-theme chance rates, which run 6–36% because broad topics have several lifelines across seven years) yields an expected **~2/10 by chance; the observed 8/10 has $p \approx 10^{-4}$** in both configurations. The gold-list performance is not an artifact of theme abundance.

Statistical robustness. Two further checks harden the detector. **Overdispersion:** weekly formation counts are strongly super-Poisson (negative-binomial dispersion $\alpha = 3.75$, method of moments on training weeks only), so a NB-variance variant was run: it is the high-precision configuration (102 lifelines vs 401; 7/10 detected at median +0 weeks; chance rate 0.9/10, $p \approx 0$), with the Poisson-floor default as the high-recall configuration — the gold-list conclusion holds under both. **Window sensitivity:** every cell of a 3×3 grid (recent 2/4/8 weeks × baseline 13/26/52

weeks) detects 7–8/10 with placebo $p < 4 \times 10^{-4}$; shorter windows detect slightly later, longer slightly earlier. Detection is not a window artifact.

Lifeline-level precision audit (supervisor-requested). The placebo establishes week-level precision; a blind audit measures the discovery stream itself. Every EMERGING lifeline was packaged with its members, birth month and provenance headlines (top articles with opening text; no detector scores) and classified REAL / TEMPLATE / INCOHERENT by a locally hosted Qwen2.5-14B judge, validated by 11/11 known controls and 93% binary agreement with a 14-pack human-labelled subset. Results: **lenient 37.9% REAL** [Wilson 95% 32.5–43.7], strict 42.9%, NB 37.1% (with the *highest* template share, 44% — event-level specificity and stream purity are distinct properties), and a new **template-filtered arm at 44.7%** [38.6–50.9]: removing template-fingerprinted *articles* (25.1% of corpus) from the detector substrate keeps every real discovery (109 vs 107) and the full gold-list result (8/10, –1 wk, $p = 5 \times 10^{-5}$) while cutting template lifelines from 86 to 54. Structural cohesion alone cannot certify quality — boilerplate binds *denser* than real news — so the semantic (judged) layer is the primary instrument; post-hoc structural filters saturate near 50% precision.

8. Limitations

- **Extractor cutoff.** Hindsight is excluded for link prediction (above) but causal-impact relations carry measurable entity-dependence.
- **US-centricity.** Thin country/regulator relations (GPE 6.7% of edges).
- **Fragmentation inflates novelty (quantified).** Surface-form variants (Apple vs Apple Inc) make some recurring relationships look novel. A crude resolution (corporate-suffix merge, 46k to 38k entities) cuts test novelty 72.4% to 68.4% and lifts recurrence MRR ~9% — so fragmentation is real but secondary, and forecasting-difficulty survives (a lower bound; full resolution would shift it further).
- **Validation scope.** Link-prediction MRRs carry 95% paired-bootstrap CIs and RE-GCN is averaged over 3 seeds; raw (unfiltered) ranking still deflates absolute MRRs relative to the filtered TKG convention, and Hits@K under the mean-rank tie convention is quantized for tie blocks straddling K (MRR unaffected). The query bootstrap treats test directions as independent — direction pairs and week clustering make the CIs somewhat narrow, which is why small gaps are additionally checked against training-seed replications (Section 5). Event recovery is scored systematically (18-event gold list, 94% burst-recall) rather than by anecdote, with the theme-level evaluation of Section 7 as the stricter, birth-window-anchored test.
- **Corpus duplication (measured).** Syndicated duplicate bodies (10.5% of 200k, 24.3% of 600k) inflate per-mention edge weights and core membership; all headline conclusions were re-verified on a deduplicated rebuild (Section 6). Exact-hash deduplication is a lower bound — near-duplicates are not caught.
- **Scale (now addressed, Section 6).** The 600k rebuild confirms denser sampling is the lever — it cut novelty 72% to 64% and lifted forecasting ~18%; residual novel-edge difficulty is the remaining ceiling.

9. Next steps

- Per-theme false-discovery audit: hand-verify a sample of detected EMERGING lifelines that match no gold event (the week-level placebo control of Section 7 is done; this is the lifeline-level complement).
- Entity-resolution pass to quantify its lift on link prediction — the next density lever now that the 600k rebuild (Section 6) is complete.
- Broaden the static-embedding baselines (e.g. RotatE / TUCKER via a KGE library) to confirm ComplEx is representative, keeping the strict chronological PIT evaluation rather than a transductive split.

10. Deliverables & reproducibility

All code is fixed-seed and resumable. Substrate: `build_subset / extract / assemble / build_core`. Dynamics (dynamics/): `loader, eda, velocity, burst (Kleinberg + CUSUM), linkpred, complex_kge, regcn, text_embed, bootstrap_ci, event_validate, themes, themes_eval`; a `KG_CORE_PATH` env switch points the same code at any core (200k / 600k / 600k-deduplicated). The 200k and 600k full KGs, denoised and deduplicated cores, raw triplet stores, subsets, and entity embeddings are archived locally; the 600k results are consolidated in `results_600k/` and the methods + literature review in an accompanying document.